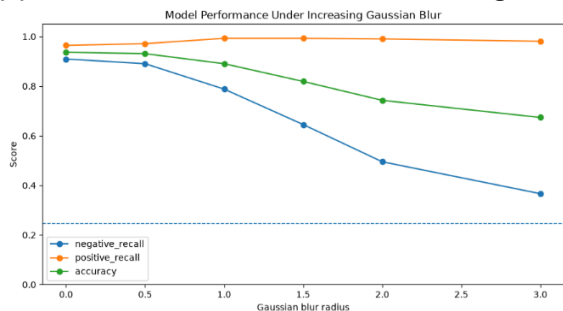
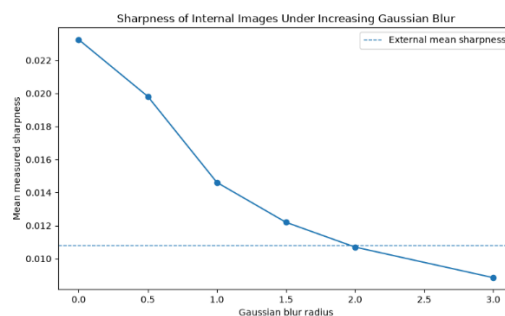


## Assessment 1 : Project 4 - AI Auditing - Report

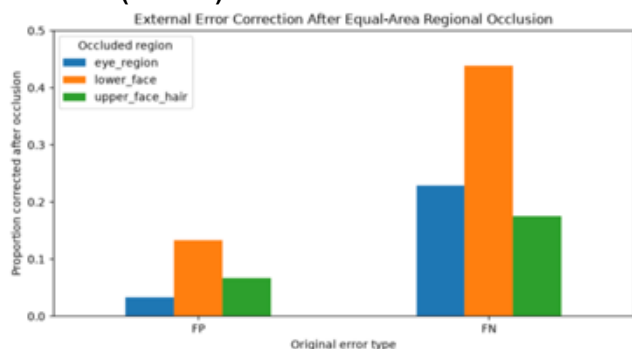
**Figure 1: Controlled sharpness perturbation (Case 1)**  
**(a) Model Performance Under Increasing Gaussian Blur**



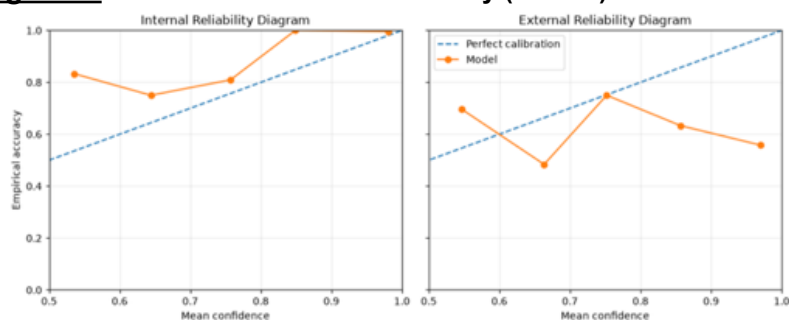
**(b) Performance and Sharpness under Increasing Gaussian Blur**



**Figure 2: Equal Area Regional Occlusion Error Correction (Case 2)**



**Figure 3: Internal vs External Reliability (Case 3)**



### Case 1 - Clean-Shaven Classifier

**Diagnosis:** The deployment failure is primarily a sharpness-related feature distribution shift exposing texture dependency. The classifier appears to rely on fine facial-hair texture learned from studio-quality images (train/val and internal test set); reduced sharpness in mobile images (the external test set) suppresses this feature and systematically biases non-clean-shaven faces toward the clean-shaven class. (ref Notebook s12.)

**Evidence.** Performance fell from 93.69% to 61.89% (internal vs external accuracy), but the degradation was highly asymmetric: positive recall remained 98.92%, while negative-class recall collapsed from 90.95% to 24.86% (ref s7–8). Dataset inspection ruled out class imbalance and resolution mismatch: both test sets were balanced 50/50 and all images were 178×218.

External images showed substantially lower measured sharpness (0.0108 vs 0.0233), while contrast remained similar (ref s9.2), and true-negative mean P(+) shifted from 0.110 to 0.682 (ref s9.1). Controlled Gaussian blur reproduced the same asymmetric failure: **Figure 1(a)** shows negative recall progressively collapsing while positive recall remains approximately 98–99% (ref s9.4). At radius 2.0, negative recall fell to 49.52% and FPR rose to 50.48%. **Figure 1(b)** shows that this perturbation produces mean sharpness of 0.0107, almost identical to the external mean of 0.0108 (s9.4.3). Brightness alone caused much smaller degradation. CAM provides supplementary evidence in Notebook s10, with the complete evidence synthesis in s11.

**Recommendations.** Fine-tune using representative mobile-domain imagery and blur/image-quality

augmentation, particularly for non-clean-shaven examples. Validate across image-quality conditions using **negative recall and FPR**, not aggregate accuracy alone. At deployment, use a sharpness-quality check to request recapture or route degraded inputs to a manual/alternative pathway, and monitor image-quality drift in production. (*Notebook s12.*)

## Case 2 — Eyeglasses Classifier

**Diagnosis.** The deployment failure is most consistent with shortcut learning caused by spurious correlations in the development distribution. The classifier does not rely exclusively on the eye/eyeglasses region and instead uses non-target facial information that appears to have been predictive internally. This relationship does not generalise reliably to the deployment population. The supplied data contain no subgroup annotations, so the audit identifies non-target feature reliance but does not assign the shortcut to a specific demographic attribute.

**Evidence.** Accuracy fell from 98.67% internally to 71.00% externally, despite both datasets being perfectly class-balanced. External errors comprised 30 false positives and 57 false negatives, with positive recall dropping from 97.33% to 62.00%. CAM inspection showed broader facial activation during some high-confidence errors. More decisively, equal-area occlusion showed that removing the lower face corrected 43.86% of false negatives, compared with only 22.81% after removing the actual eye region (**Figure 2**). Lower-face masking also corrected more false positives than eye-region masking (13.33% vs 3.33%).

**Recommendations.** Rebalance development data to break correlations between eyeglasses and unrelated facial characteristics, evaluate performance across documented user subgroups, and investigate eye-region-focused augmentation or cropping. Before redeployment, repeat CAM and controlled occlusion tests to verify reduced dependence on non-target features.

## Case 3 — Young Classifier

**Diagnosis.** The deployment failure is caused by severe confidence miscalibration under distribution shift, specifically external overconfidence. The High-Confidence Bypass assumes that a model prediction with  $\geq 90\%$  softmax confidence has a correspondingly high probability of being correct. This assumption is approximately valid on the internal development distribution but breaks down after deployment. Consequently, the confidence score no longer provides a reliable measure of prediction risk, allowing incorrect classifications to bypass human review despite being reported with near-certain confidence.

**Evidence.** Overall accuracy fell from 97.33% internally to 59.33% externally, demonstrating a substantial generalisation failure. More importantly, calibration deteriorated sharply: Expected Calibration Error increased from 0.0362 to 0.2956. At the operational  $\geq 90\%$  bypass threshold, 243 internal predictions qualified for automatic publication and achieved 99.59% accuracy, producing only one error. In contrast, 163 external predictions exceeded the same threshold but achieved only 55.83% accuracy, meaning 72 incorrect predictions would have been automatically published without human review. The reliability analysis provides direct evidence of overconfidence (**Figure 3**): within the external 90–100% confidence bin, mean reported confidence was 96.99%, while empirical accuracy was only 55.83%, a 41.16 percentage-point calibration gap. Thus, confidence remains extremely high even when deployment accuracy collapses.

**Recommendations.** Suspend the confidence-only bypass until confidence reliability has been demonstrated on deployment-representative data. Apply an appropriate post-hoc calibration method using a separate deployment-representative validation set, then select any future automation threshold according to a predefined acceptable error rate rather than interpreting raw softmax probability literally. The underlying classification model must also be investigated because calibration can improve the reliability of confidence estimates but cannot correct the model's 59.33% external accuracy. Development data should therefore be expanded or revised to better represent deployment conditions, with external accuracy, calibration, high-confidence error rate and bypass accuracy monitored after redeployment.